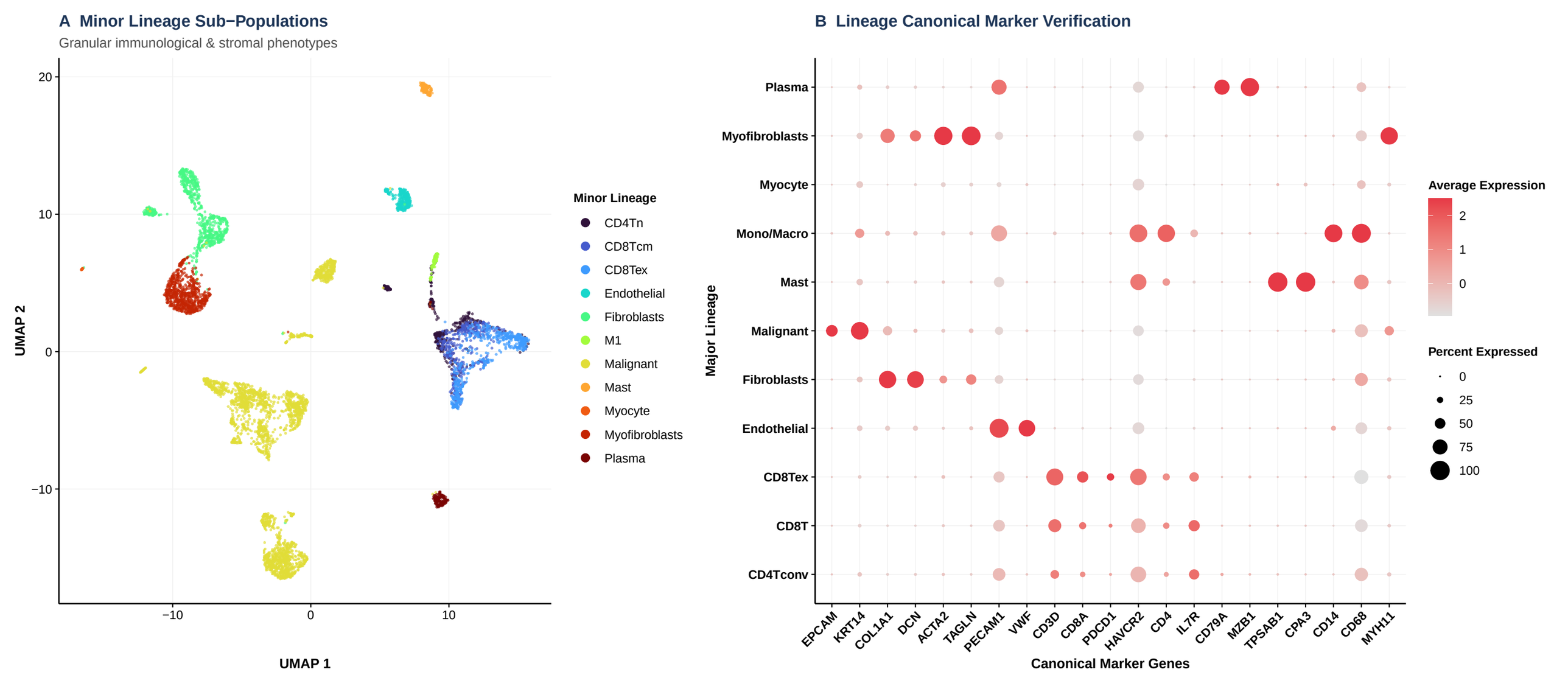

A Minor Lineage Sub−Populations
Granular immunological & stromal phenotypes
UMAP 2
UMAP 1
20
10
0
−10
−10
0
10
Minor Lineage
CD4Tn
CD8Tcm
CD8Tex
Endothelial
Fibroblasts
M1
Malignant
Mast
Myocyte
Myofibroblasts
Plasma
B Lineage Canonical Marker Verification
Major Lineage
Plasma
Myofibroblasts
Myocyte
Mono/Macro
Mast
Malignant
Fibroblasts
Endothelial
CD8Tex
CD8T
CD4Tconv
EPCAM
KRT14
COL1A1
DCN
ACTA2
TAGLN
PECAM1
VWF
CD3D
CD8A
PDCD1
HAVCR2
CD4
IL7R
CD79A
MZB1
TPSAB1
CPA3
CD14
CD68
MYH11
Canonical Marker Genes
Average Expression
2
1
0
Percent Expressed
0
25
50
75
100